

AI-Powered Campus Assistant for Students

Lenovo Leap AI Capstone Project

GCOEN Campus Assistant is an AI-inspired web application using Natural Language Processing concepts and automated response mechanisms to provide students with quick access to academic resources, notices, events, placement information and campus support.

Submitted By

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SDG Selected: SDG 4 – Quality Education

1. Selected SDG and Reason for Selection:

SDG Selected

United Nations Sustainable Development Goal 4 (Quality Education)

About SDG 4

SDG 4 aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The goal focuses on improving access to educational resources, reducing barriers to learning, and enhancing educational outcomes through innovation and technology.

Reason for Selection

Education plays a vital role in personal and professional development. In colleges, students often struggle to find important academic information, notices, study materials, event updates, and placement-related resources. Due to scattered information sources, students spend significant time searching for information instead of focusing on learning.

To address this issue, an AI-powered Campus Assistant was developed to provide students with instant access to academic resources and college-related information through a single platform.

2. Problem Statement:

Problem Description

Students frequently face difficulties in accessing important college-related information such as:

- Academic schedules
- Examination notifications

- Study materials
- Placement opportunities
- Event announcements
- Faculty information

The information is often distributed across multiple websites, WhatsApp groups, notice boards, and social media platforms.

Why Does This Problem Exist?

- Lack of a centralized information system
- Information scattered across different sources
- Difficulty in searching for relevant resources
- Limited student awareness about available resources

Who Is Affected?

- Undergraduate students
- Newly admitted students
- Faculty members
- College administration

Why Is It Serious?

Students may miss important deadlines, examination notices, event registrations, internship opportunities, and placement updates.

Consequences If Not Solved

- Reduced student productivity
- Communication gaps
- Missed educational opportunities
- Increased confusion among students

3. Proposed Solution:

Project Name

GCOEN Campus Assistant

Project Description

GCOEN Campus Assistant is an AI-powered web application designed to provide students with quick and easy access to college-related information through a user-friendly interface.

The platform acts as a virtual assistant capable of answering student queries and providing educational resources instantly.

How the System Works

1. Student opens the web application.
2. User enters a query or selects an option.
3. AI processes the request.
4. Student receives accurate information instantly.

How It Solves the Problem

- Centralizes academic information
- Reduces time spent searching
- Improves communication
- Provides quick access to educational resources

Target Users

- Students
- Faculty members
- College administration

4. Project Features:

1. AI Chat Assistant

An intelligent chatbot that answers student queries related to academics, resources, events, and campus information.

2. Academic Resource Access

Provides quick access to notes, study materials, reference books, and learning resources.

3. Event Information

Displays upcoming college events, workshops, seminars, and technical activities.

4. Notice Management

Provides centralized access to important notices and announcements.

5. User-Friendly Interface

Responsive and simple interface that works across desktop and mobile devices.

6. Quick Search Functionality

Allows users to search and locate information instantly.

5. Technology Stack:

Frontend Technologies

- HTML5
- CSS3
- JavaScript

Artificial Intelligence Technologies

- Generative AI
- Natural Language Processing (NLP)

Development Tools

- Visual Studio Code
- GitHub

Hosting Platform

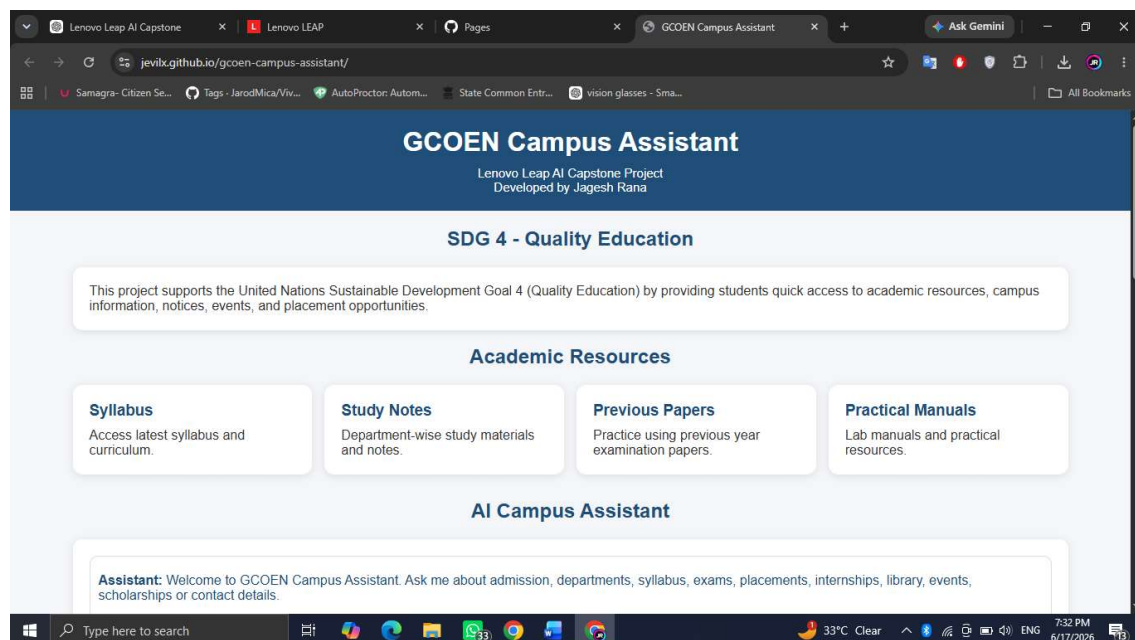
- GitHub Pages

Version Control

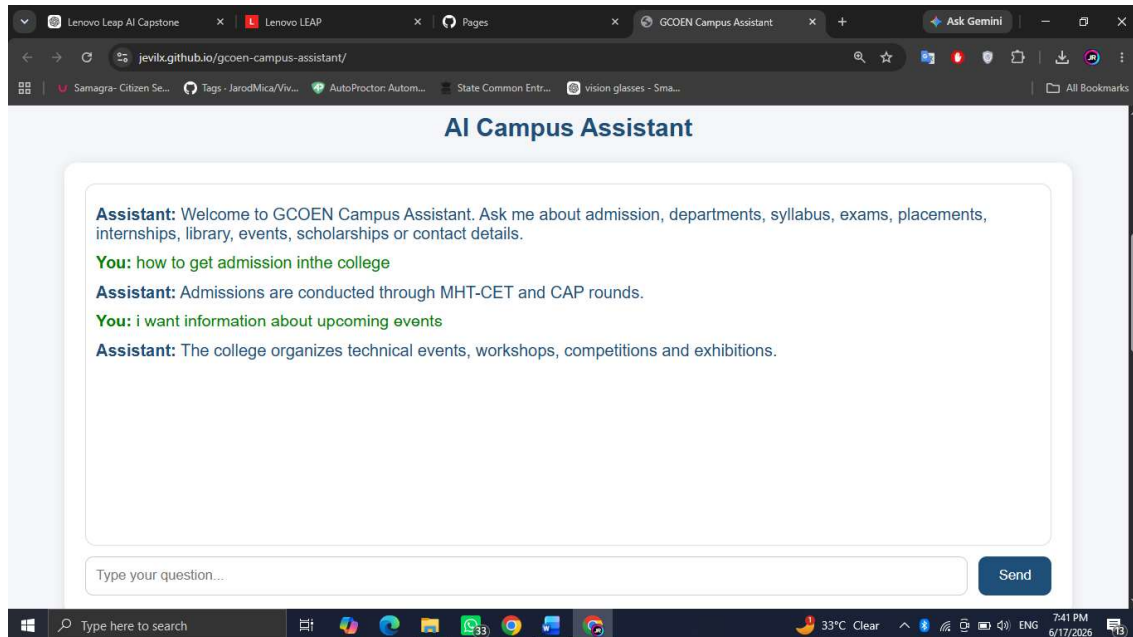
- Git

6. Project Screenshots (Proof of Work):

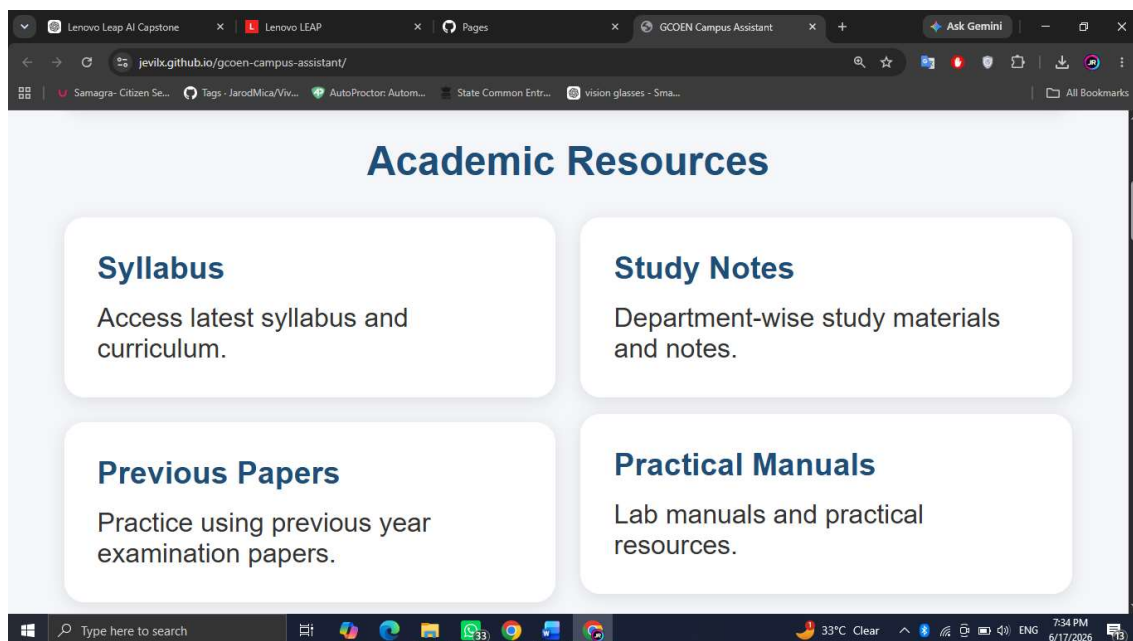
Screenshot 1: Homepage



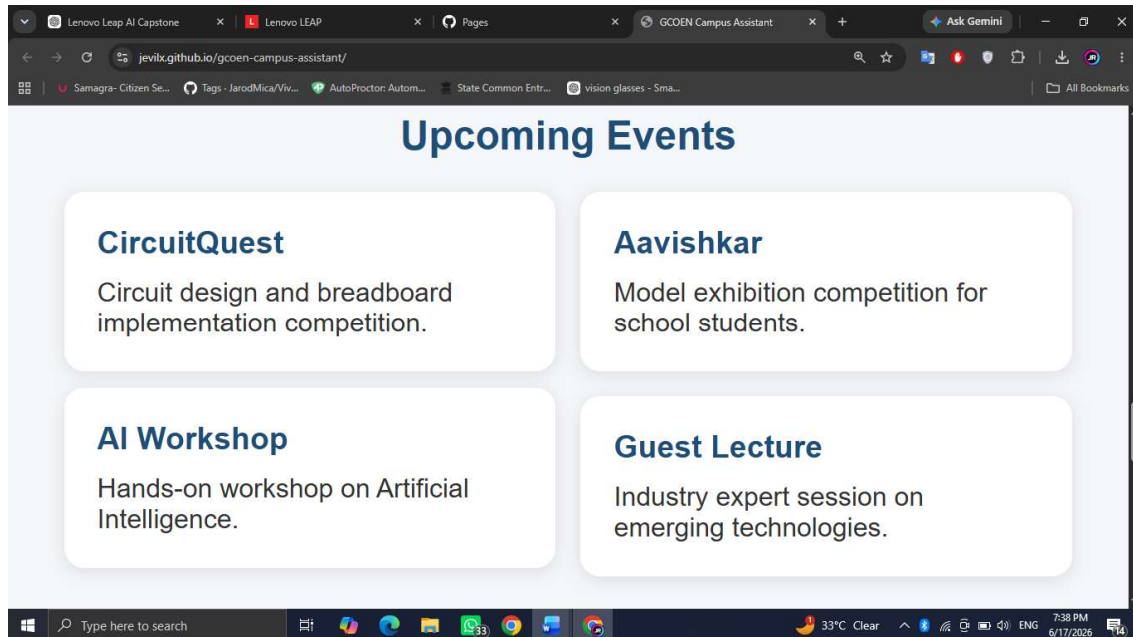
Screenshot 2: AI Chat Interface



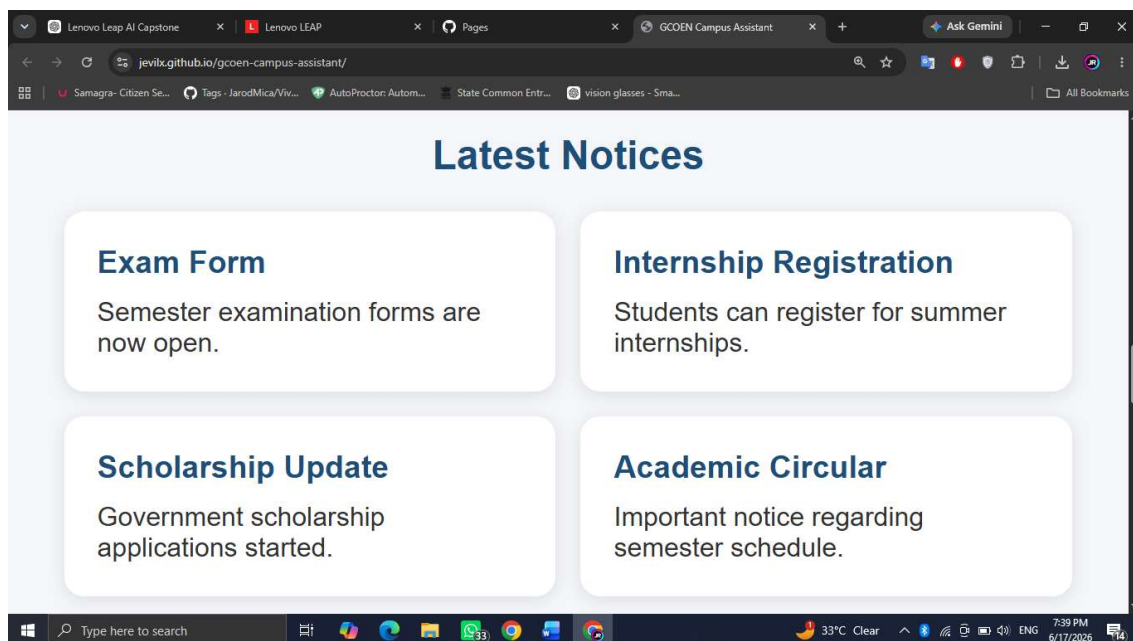
Screenshot 3: Resource Section



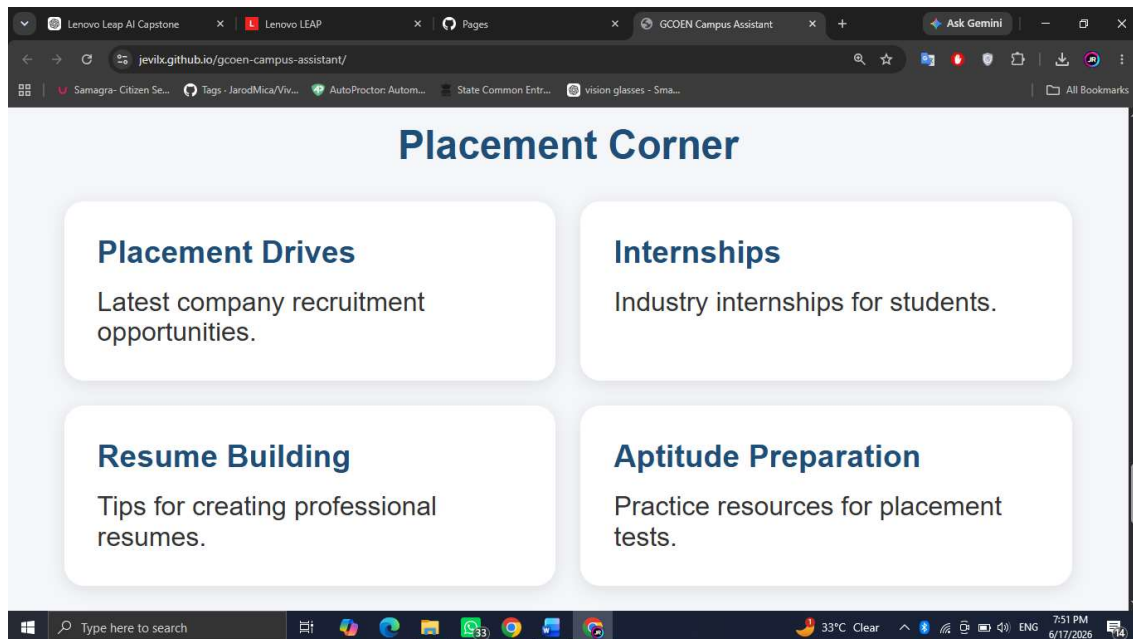
Screenshot 4: Event Information Section



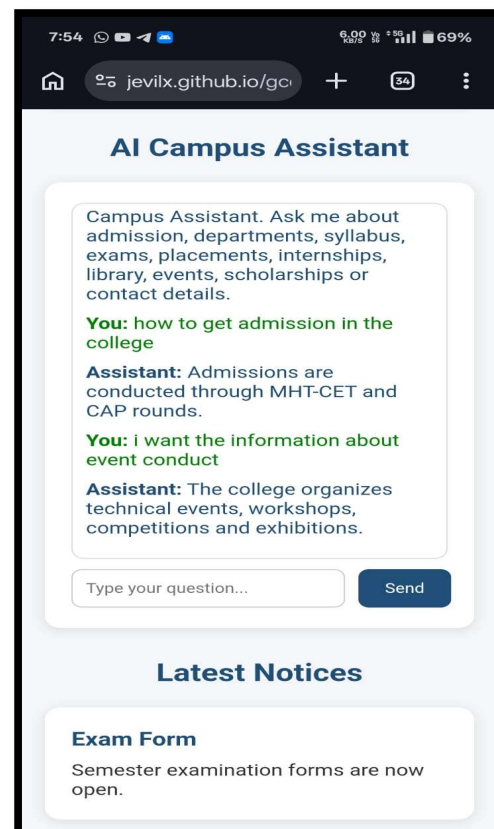
Screenshot 5: Notice Dashboard



Screenshot 6: Placement Corner



Screenshot 7: Mobile Responsive View



7. Open Source Repository:

GitHub Repository

<https://github.com/jevilx/gcoen-campus-assistant>

Live Project Link

<https://jevilx.github.io/gcoen-campus-assistant/>

The complete source code, project files, and documentation are available in the GitHub repository.

8. Future Scope:

The project can be further enhanced through the following improvements:

Voice-Based Assistant

Enable voice interaction for hands-free access.

Mobile Application

Develop Android and iOS applications.

Multilingual Support

Support multiple regional languages.

Personalized Dashboard

Provide customized recommendations based on student interests.

ERP Integration

Connect with college management systems for real-time updates.

AI Recommendation System

Suggest study materials, events, internships, and opportunities based on student profiles.

Conclusion:

The AI-Powered Campus Assistant successfully addresses the problem of fragmented educational information by providing a centralized platform for students. The system improves accessibility, saves time, enhances communication, and supports the objectives of SDG 4: Quality Education.

The project demonstrates how Artificial Intelligence can be used effectively to improve educational experiences and contribute toward sustainable development goals.

References

1. United Nations Sustainable Development Goals
2. GitHub Documentation
3. HTML, CSS and JavaScript Documentation
4. GitHub Pages Documentation
5. Artificial Intelligence and NLP Learning Resources